CRLB (continued)

We continue with the last example of Lecture 13, viz.,

Ex 3 $X_1, X_2, \ldots, X_n$ are iid $N(\theta, 1)$, $\theta \in \mathbb{R}$.

But now, $g(\theta) = \theta^2$.

As before, $I(\theta) = 1$, so that $CRLB = \frac{(2\theta)^2}{n} = \frac{4\theta^2}{n}$.

$\because E(\bar{X}^2) = V(\bar{X}) + E^2(\bar{X}) = \frac{1}{n} + \theta^2$, we easily get the UMVUE of $g(\theta) = \theta^2$, as $\delta(\underset{\sim}{X}) = \bar{X}^2 - \frac{1}{n}$,

Note $\bar{X}$ is the CSS and $E(\bar{X}^2 - \frac{1}{n}) = \theta^2$, so by the Lehman Scheffe theorem, ~~$\delta(\bar{X})$ is~~ $\delta(\underset{\sim}{X}) = \bar{X}^2 - \frac{1}{n}$ is the UMVUE.

The UMVUE has min^{m} var. among all unbiased estimators.

$V(\bar{X}^2 - \frac{1}{n}) = Var(\bar{X}^2) = E(\bar{X}^4) - E^2(\bar{X}^2)$

For calculation of the 4^{th} moment, use transformation $Z = (\bar{X} - \theta)\sqrt{n} \sim N(0,1)$, so that $E(Z) = E(Z^3) = 0$, $E(Z^2) = 1$, and $E(Z^4) = 3$. Hence,

$$V(\bar{X}^2 - \tfrac{1}{n}) = Var(\bar{X}^2) = E(\bar{X}^4) - E^2(\bar{X}^2)$$
$$= \left(\theta^4 + \frac{6\theta^2}{n} + \frac{3}{n^2}\right) - \left(\theta^2 + \frac{1}{n}\right)^2 = \frac{4\theta^2}{n} + \frac{2}{n^2} > \frac{4\theta^2}{n}, \text{ the CRLB!}$$

So, this is an example where the UMVUE of $g(\theta)$ does not the CRLB.

Effect of Reparametrization on the Fisher information, $I(\theta)$.

In the above example, if instead of $I(\theta)$, we worked with $I(\theta^2)$, this is logical since the parameter of interest is $g(\theta) = \theta^2$.

The problem $\theta = h(\xi)$ for real valued, continuous function. To obtain $I(\xi)$ in terms of $I(\theta)$ (a reason could be calculating $I(\theta)$ is easier).

Another example: $X_1, \ldots, X_n$ iid $N(0, \sigma^2)$, $\sigma > 0$.

$\sigma^2 = h(\sigma)$, to obtain $I(\sigma)$ in terms of $I(\sigma^2)$

$$\underline{I(\xi)} = E\left[\frac{\partial}{\partial \xi} \ln f_{h(\xi)}(x)\right]^2 = E\left[\frac{\partial \theta}{\partial \xi} \cdot \frac{\partial}{\partial \theta} \log f_\theta(x)\right]^2$$

$$= \left(\frac{\partial \theta}{\partial \xi}\right)^2 E\left[\frac{\partial}{\partial \theta} \ln f_\theta(x)\right]^2 = \underline{(h'(\xi))^2 \, I(\theta)}.$$

However, $\because \; g'(\theta) = \frac{\partial}{\partial \theta}(g(\theta)) = \frac{\partial \xi}{\partial \theta} \cdot \frac{\partial}{\partial \xi}(g^*(\xi))$, $\quad g^*(\xi) = g(h(\xi))$.

$$(g'(\theta))^2 = \frac{1}{(h'(\xi))^2} (g^{*\prime}(\xi))^2.$$

$\therefore$ the CRLB of any unbiased estimator of ξ, can be given by,

$$\underline{\mathrm{Var}(\delta^*(\underline{x})) \geq \frac{(g^{*\prime}(\xi))^2}{I(\xi)}}.$$

__Ex__ Try to obtain CRLB's of unbiased estimators of σ^2 and σ, for the $N(0, \sigma^2), \sigma > 0$ population.

__Note__: We've considered the case for a real valued (scalar) θ only. The multi-parameter case is not considered in this course.

__Next__: Method of Moments Estimator.
and Minimum Mean Square Error Estimator.